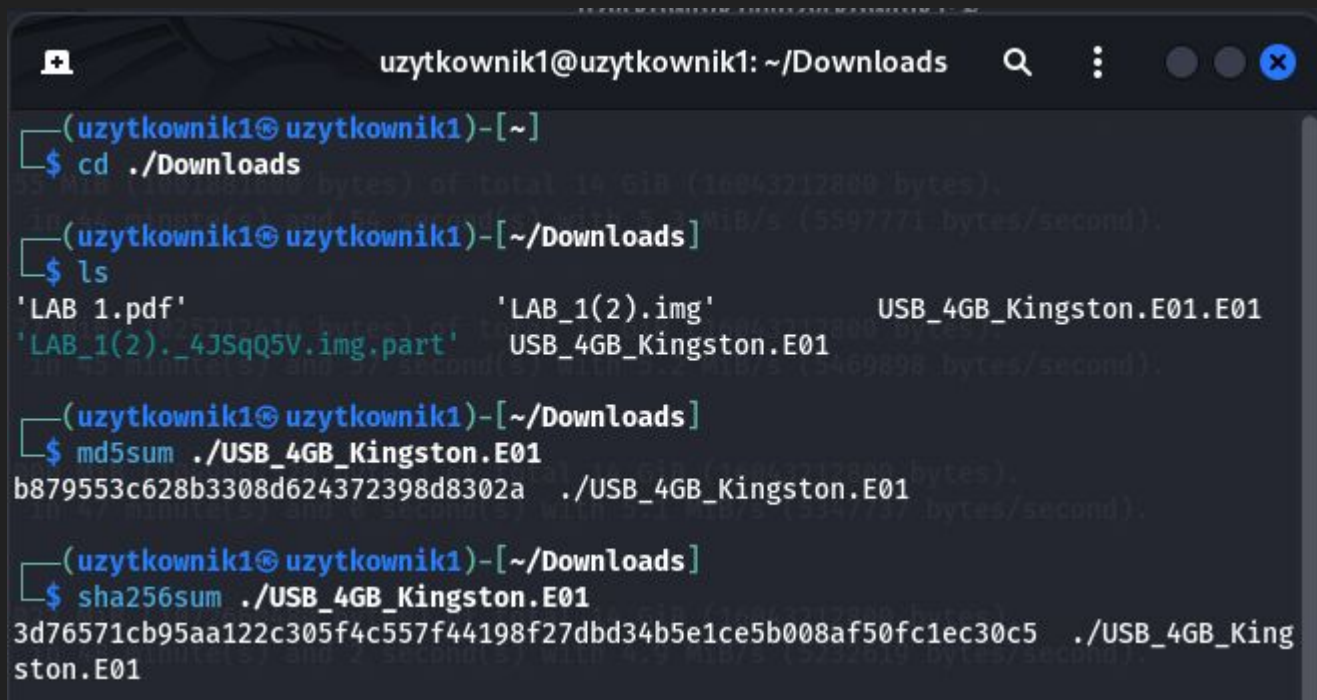


Laboratoria nr 1

Bartłomiej Nałódka

Zad 2. - wartości funkcji skrótów - jak widać na screenie



```
(uzytkownik1@uzytkownik1)-[~]
$ cd ./Downloads

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ ls
'LAB 1.pdf'                'LAB_1(2).img'                USB_4GB_Kingston.E01.E01
'LAB_1(2)._4JSqQ5V.img.part'  USB_4GB_Kingston.E01

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ md5sum ./USB_4GB_Kingston.E01
b879553c628b3308d624372398d8302a  ./USB_4GB_Kingston.E01

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ sha256sum ./USB_4GB_Kingston.E01
3d76571cb95aa122c305f4c557f44198f27dbd34b5e1ce5b008af50fc1ec30c5  ./USB_4GB_Kingston.E01
```

Niezaalokowana pamięć
znajduje się w 1. przedziale
sektorów (bo bity od 0. do
127. mieszczą się w
zakresie <0,512>).

Pliki systemowe znajdują się
w partycji 002.

Początek sektora Win95:
0000000128

Koniec: 0007581695

```
uzytkownik1@uzytkownik1: ~/Downloads
Setting up gnutls-bin (3.8.6-2) ...
Setting up libcurl4t64:amd64 (8.10.1-2) ...
Setting up libcurl3t64-gnutls:amd64 (8.10.1-2) ...
Setting up libafflib0t64:amd64 (3.7.20-2) ...
Setting up curl (8.10.1-2) ...
Setting up libtsk19t64:amd64 (4.12.1+dfsg-0kali6) ...
Setting up sleuthkit (4.12.1+dfsg-0kali6) ...
Processing triggers for libc-bin (2.40-2) ...
Processing triggers for man-db (2.12.0-1) ...
Processing triggers for kali-menu (2023.4.6) ...

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmls ./USB_4GB_Kingston.E01
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length      Description
000:  Meta      00000000000  00000000000  00000000001  Primary Table (#0)
001:  -----      00000000000  00000000127  00000000128  Unallocated
002:  000:000      00000000128  0007581695  0007581568  Win95 FAT32 (0x0c)

(uzytkownik1@uzytkownik1)-[~/Downloads]
$
```

Typ plików - FAT32

Wielkość sektora: 512 [B]

Wielkość klastra: 8192 [B]

Podane w bajtach - [B]

```
uzytkownik1@uzytkownik1: ~/Downloads
$ fsstat ./USB_4GB_Kingston.E01 -o 128
FILE SYSTEM INFORMATION
-----
File System Type: FAT32
OEM Name: MSDOS5.0
Volume ID: 0x779c953c
Volume Label (Boot Sector): USB DISK
Volume Label (Root Directory):
File System Type Label: FAT32
Next Free Sector (FS Info): 11392
Free Sector Count (FS Info): 7504624

Sectors before file system: 128

File System Layout (in sectors)
Total Range: 0 - 7581567
* Reserved: 0 - 47
** Boot Sector: 0
** FS Info Sector: 1
** Backup Boot Sector: 8
* FAT 0: 48 - 3751
* FAT 1: 3752 - 7455
* Data Area: 7456 - 7581567
** Cluster Area: 7456 - 7581567
*** Root Directory: 7456 - 7471

METADATA INFORMATION
-----
Range: 2 - 121185798
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 8192
Total Cluster Range: 2 - 473383

FAT CONTENTS (in sectors)
-----
7456-7471 (16) -> EOF
7472-7487 (16) -> EOF
7488-7503 (16) -> EOF
7552-7583 (32) -> EOF
9552-9567 (16) -> EOF
9568-9583 (16) -> EOF
```

```

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ fls -m / USB_4GB_Kingston.E01 -o 128
0|/USB DISK (Volume Label Entry)|3|r/rrwxrwxrwx|0|0|0|0|1633274396|0|0
0|/.Spotlight-V100|6|d/dr-xr-xr-x|0|0|8192|1633215600|1633274286|0|1633274286
0|/.fseventsd (deleted)|8|d/dr-xr-xr-x|0|0|8192|1633215600|1633274286|0|1633274286
0|/1|9|d/drwxrwxrwx|0|0|8192|1633215600|1633274374|0|1633274375
0|/IMG_5609.JPG|10|r/rrwxrwxrwx|0|0|5626265|1633215600|1625919170|0|1632667325
0|/. _IMG_5609.JPG (deleted)|13|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/IMG_5627.JPG|14|r/rrwxrwxrwx|0|0|4442652|1633215600|1625919414|0|1632667325
0|/. _IMG_5627.JPG (deleted)|17|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/IMG_5753.JPG|18|r/rrwxrwxrwx|0|0|5350922|1633215600|1626625912|0|1632667325
0|/. _IMG_5753.JPG (deleted)|21|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/IMG_6002.JPG|22|r/rrwxrwxrwx|0|0|2642384|1633215600|1627153216|0|1632667325
0|/. _IMG_6002.JPG (deleted)|25|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/IMG_8064.JPG|26|r/rrwxrwxrwx|0|0|6511470|1633215600|1628355454|0|1632667325
0|/. _IMG_8064.JPG (deleted)|29|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/text2.rar|30|r/rrwxrwxrwx|0|0|138|1633215600|1632667466|0|1632667499
0|/. _text2.rar (deleted)|32|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/. _1 (deleted)|34|r/rr-xr-xr-x|0|0|4096|1633215600|1633274378|0|1633274376
0|/$MBR|121185795|v/v-----|0|0|512|0|0|0|0
0|/$FAT1|121185796|v/v-----|0|0|1896448|0|0|0|0
0|/$FAT2|121185797|v/v-----|0|0|1896448|0|0|0|0
0|/$OrphanFiles|121185798|V/V-----|0|0|0|0|0|0|0

```

Opcja -m / wskazuje na główny katalog w pendrive. Widzimy jego zawartość. Aby wyświetlić wszystkie pliki należałoby dodać jeszcze opcję -r.

```
—(uzytkownik1@uzytkownik1)-[~/Downloads]
```

```
0|/1/USB DISK (Volume Label Entry)|3|r/rrwxrwxrwx|0|0|0|0|1633274396|0|0
```

0|/1/.Spotlight-V100/Store-V2|262|d/drwxrwxrwx|0|0|8192|1633215600|1633274286|0|1633274286

0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAEF70|520|d/drwxrwxrwx|0|0|16384|1633215600|1633274286|0|1633274286

0|1/2 Spotlight-V100/Store-V2/6F621758-5262-4AB0-BAD2-Q6DA8EDBBAEF70/psid.db[1541|r/r/rwx/rwx/rwx|0|0|8192|1633215600|1633274396|0|1633274368

0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAEF70/tmp.|ion|1543|r/rwxrwxrwx|0|0|0|1633215600|1633274368|0|1633274368

```
0|/1/ Spotlight-V100/Store-V2/GF6E21758-5262-4AB0-BAD2-96DABEDBAEF70/ion_created|1545|r/rwxrwxrwx|0|0|0|16332215600|16332274368|0|16332274368
```

0|/1/ Spotlight-V100/Store-V2/6F621758-5262-4AB0-BAD2-96DA8EDBAE70/tmp_Cab|1547|r/rwxrwxrwx|0|0|0|1633215600|1633274368|0|1633274368

```
01/1/ Spotlight-V100/Store-V2/6E621758-5262-4AB0-BAD2-90DA8EDBAF70/ cmh.Cab 1547 1171W1W1W1W1 0 0 0 1053215000 1053215000 0 1053215000
```

0/1/1/Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-06DA8EDBA170/cal.created 1549 1/11WX1WX1WX 0 0 0 1053215000 1053274308 0 1053274308
0/1/1/Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-06DA8EDBA170/indexState 1551 r/rrrrrrrrrrrr 0 0 0 1633215600 163327306 0 163327368

```
0/1/Spotlight-V100/Store-V2/6E621758-5262-4AB0-BAD2-90D8AEDBBAF7/0/indexState/155137/11Wx11Wx1Wx/0/0/20/1053215000/1053214330/0/1053274300
0/1/Spotlight-V100/Store-V2/6E621758-5262-4AB0-BAD2-06D8AEDBBAF7/0/indexArrows/155137/r/11Wx11Wx1Wx/0/0/20/1053215000/1053215600/0/1633277372/0/1633277372
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
0	1	2	3	4	5	6	7	8	9	10	11	12																																																																																							

```
0/1/1:Spotlight-V100/Store-V2/6F621758-3262-4AB9-BAD2-96DA8EDBAF70/0_IndexDataIndex1555/1/1/WX1WX1WX/0/0/10/1033210000/103321432/0/1033214300
```

0/1/1	Spotlight-V100/Store-V2/0/0/2	3208-3202-4AB9	BAD2-90DA8CDBAF70/0	11104ABigDates	1556	1/1/1W1W1W1W1	0/0/0	10532213000	10532243570	10532243508
0/1/1	Spotlight-V100/Store-V2/6/6/2	3262-4AB9	BAD2-06DA8CDBAF70/0	indexGroups	1560	1/1/1W1W1W1W1	0/0/0	16322215600	16322273720	16322273660

0/1/1/Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/0.indexGroups 1560 1/1FWXFWXFWX 0 0 3277 1633215600 1633274370 0 1633274368
0/1/1/Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/0.indexPositions 1563 1/1FWXFWXFWX 0 0 4 1633315600 1633374370 0 1633374368

```
0/1/.Spotlight-V100/Store-V2/6F621D58-3262-4AB9-BAD2-96DA8EDBAF70/0.indexPOSITIONS15631/r/rwxrwxrwx|0|4|16332213000|1633224372|0|1633224372|
0/1/.Spotlight-V100/Store-V2/6F621D58-3262-4AB9-BAD2-96DA8EDBAF70/0.indexPOSITIONS15661/r/rwxrwxrwx|15666|16332215600|1633224372|0|1633224372|
```

```
0/1/.Spotlight-V100/Store-V2/6f621758-3262-4AB9-BAD2-96DA8EDBBAF0/0_indexPostings|1366|r/rwxrwxrwx|0|0|122|1633213300|163324372|0|163324372|
```

```
01/17.Spotlight-V100/Store-V2/6F621F35-5262-4AB9-BAD2-96DA8EDBAF70/0.IndexDirectory|1369|r/rfwwrxwrxw|0|0|2056|1633215600|163321437Z|0|163321437Z
```

```
01/1/Spotlight-V100/Store-V2/6F621750-5262-4AB9-BAD2-96DA8EDBAF70/0.IndexCompactDirectory/15/2/F/rrwxrwxrwx/0/0/8/1633215600/1633214372/0/1633214372/
```

```
01/1/.Spotlight-V100/Store-V2/6f621f5c-5262-4AB9-BAD2-96DA8EDBAF/0/0.shadowIndexHead/15/5/r/rfwrwxrwx/0/0/4096/1633215600/16332/43/2/0/16332/43/3
```

```
0|/1/.Spotlight-V100/Store-V2/6f6217515262-4AB9-BAD2-96bDA8EDBAF70/live.0.indexUpdates|1578|r/rwxrwxrwx|0|0|8799|1633215600|1633274396|0|1633274397|
```

0/1/.Spotlight-V100/Store-V2/6f6217515262-4AB9-BAD2-96DA8EDBAF/0/_MP0CM~1.SHA (deleted)	1579	r/rwxrwxrwx	0	0	1088	1633215600	1633274372	0	16332743
---	------	-------------	---	---	------	------------	------------	---	----------

0/1/.Spotlight-V100/Store-V2/6F621758-3262-4AB9-BAD2-96DA8EDBAF70/0.directoryStoreFile|582|r/rwxrwxrwx|0|0|65536|1633215600|1633274372|0|1633274372

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/0.directoryStoreFile.shadow|1586|r/rrwxrwxrwx|0|0|1088|1633215600|1633274372|0|163
```

```
0|1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexHead|1589|r/rrwxrwxrwx|0|0|4096|1633215600|1633274396|0|1633274368
```

```
0|/1/.Spotlight-V100/Store-V2/6F621755-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexIds|1592|r/rrwxrwxrwx|0|0|32768|1633215600|1633274396|0|1633274368
```

```
0|/1/.Spotlight-V100/Store-V2/6F621755-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexBigDates|1595|r/rwxrwxrwx|0|0|16384|1633215600|1633274396|0|163327436
```

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexGroups|1598|r/rrwxrwxrwx|0|0|3277|1633215600|1633274396|0|1633274368
```

```
0|/1/.Spotlight-V100/Store-V2/6F621755-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexPostings|1601|r/rwxrwxrwx|0|0|4096|1633215600|1633274370|0|1633274368
```

0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexTermIds|1604|r|rrwxrwxrwx|0|0|8192|1633215600|1633274368|0|1633274368

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexPositions|1607|r/rrwxrwxrwx|0|0|4096|1633215600|1633274368|0|1633274368
```

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexPositionTable|1610|r/rrwxrwxrwx|0|0|8192|1633215600|1633274368|0|16332
```

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexDirectory|1613|r/rrwxrwxrwx|0|0|8224|1633215600|1633274370|0|163327436
```

```
0|1/.Spotlight-V100/Store-V2/6F621758-3262-4AB9-BAD2-96DA8EDBAF70/live.0.indexCompactDirectory|1617|r/rrwxrwxrwx|0|0|1024|1633215600|1633274370|0|1633274368
```

```
0|/1..Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.indexArrays|1620|r/rrwxrwxrwx|0|0|65536|1633215600|1633274368|0|1633274368
```

0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/store.updates|1622|r/rwxrwxrwx|0|0|8|1633215600|1633274396|0|1633274397

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/_IVE~~88.IND (deleted)|1623|r/rwxrwxrwx|0|0|0|1633215600|1633274372|0|1633274372
```

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/live.0.directoryStoreFile|1626|r/rrwxrwxrwx|0|0|65536|1633215600|1633274368|0|1633274368
```

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/store.db|1627|r/rrwxrwxrwx|0|0|36864|1633215600|1633274372|0|1633274368
```

0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/.store.db|1629|r/rr-xr-xr-x|0|0|36864|1633215600|1633274396|0|1633274368

```
0|/1/.Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/dbStr-1.map.header|1632|r|rrwxrwxrwx|0|0|56|1633215600|1633274396|0|1633274368
```

0		/	/	.	S	p	o	t	l	i	g	h	-	V	1	0	0	/	S	t	o	r	e	-	V	2	/	6	F	6	2	1	7	5	8	-	5	2	6	2	-	4	A	B	9	-	B	A	D	2	-	9	6	D	A	8	F	D	B	A	F	7	0	/	d	b	S	t	r	-	1	.	m	a	p	.	d	a	t	a		1	6	3	5		r	/	r	r	w	x	r	w	x	r	w	x		0		0		4	0	9	6		1	6	3	3	2	1	5	6	0		1	6	3	3	2	7	4	3	9	6		0		1	6	3	3	2	7	4	3	6	8
---	--	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	--	---	---	---	---	--	---	---	---	---	---	---	---	---	---	---	---	---	--	---	--	---	--	---	---	---	---	--	---	---	---	---	---	---	---	---	---	--	---	---	---	---	---	---	---	---	---	---	--	---	--	---	---	---	---	---	---	---	---	---	---

```
0/1/. Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAF70/dbStr-1.map.offsets 1638 | r | rrrwxrwxrwx | 0 | 0 | 4096 | 1633215600 | 1633274396 | 0 | 1633274368
```

0	/1/ Spotlight-V100/Store-V2/6F621758-5262-4AB9-BAD2-96DA8EDBAE70/dhStr-1.map_buckets	1641	r r r w x r w x r w x	0	0	4096	16332215600	16332274396	0	16332274368
---	--	------	-----------------------	---	---	------	-------------	-------------	---	-------------

Na poprzednim slajdzie wypisałem naprawdę wszystkie pliki

Właśnie z użyciem opcji -r. Jak widać, wyniki są mało czytelne.

1. Nr sprawy: 001
2. Nazwa twórcy obrazu: Kali
3. Data utworzenia: 03.10.2021
4. Nr seryjny i nazwa modelu: 0D7117891080, USB DISK 2.0
5. Format plików: EnCase 6
6. Metoda kompresji: deflate
7. Pełna wielkość w bajtach: 3881828352
8. Poziom kompresji wskazany przy tworzeniu pliku: good (fast)

```
uzytkownik1@u
0|/1/.IMG_6002.JPG (deleted)|25|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/1/IMG_8064.JPG|26|r/rrwxrwxrwx|0|0|6511470|1633215600|1628355454|0|1632667325
0|/1/.IMG_8064.JPG (deleted)|29|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/1/text2.rar|30|r/rrwxrwxrwx|0|0|138|1633215600|1632667466|0|1632667499
0|/1/.text2.rar (deleted)|32|r/rr-xr-xr-x|0|0|4096|1633215600|1633274374|0|1633274375
0|/1/.1 (deleted)|34|r/rr-xr-xr-x|0|0|4096|1633215600|1633274378|0|1633274376
0|/1/$MBR|121185795|v/v-----|0|0|512|0|0|0|0
0|/1/$FAT1|121185796|v/v-----|0|0|1896448|0|0|0|0
0|/1/$FAT2|121185797|v/v-----|0|0|1896448|0|0|0|0
0|/1/$OrphanFiles|121185798|V/V-----|0|0|0|0|0|0|0
10843212800 bytes)
10843212800 bytes/sec

(uzytkownik1@ uzytkownik1) - [~/Downloads]
$ ewfinfo ./USB_4GB_Kingston.E01
ewfinfo 20140814

Acquiry information
Case number: 001
Examiner name: Kali
Evidence number: 001
Acquisition date: Sun Oct 3 15:31:05 2021
System date: Sun Oct 3 15:31:05 2021
Operating system used: Linux
Software version used: 20140807
Password: N/A
Model: USB DISK 2.0
Serial number: 0D7117891080

EWF information
File format: EnCase 6
Sectors per chunk: 64
Error granularity: 64
Compression method: deflate
Compression level: good (fast) compression

Media information
Media type: removable disk
Is physical: yes
Bytes per sector: 512
Number of sectors: 7581696
Media size: 3.6 GiB (3881828352 bytes)

Digest hash information
MD5: 5df8f604967c556c810d21dd664ceae4
```

```
(uzytkownik1@ uzytkownik1) - [~/Downloads]
$ ss
```

```

(uzytownik1@uzytownik1)-[~/Downloads]
$ mmls ./LAB_1.img
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End      Length    Description
000:  Meta      0000000000 0000000000 0000000001 Safety Table
001:  -----  0000000000 0000002047 0000002048 Unallocated
002:  Meta      0000000001 0000000001 0000000001 GPT Header
003:  Meta      0000000002 0000000033 0000000032 Partition Table
004:  000      0000002048 0000104447 0000102400 fat16
005:  001      0000104448 0000309247 0000204800 fat32
006:  002      0000309248 0000718847 0000409600 ntfs
007:  003      0000718848 0001058815 0000339968 ext4
008:  004      0001058816 0001091583 0000032768 swap
009:  005      0001091584 0001173503 0000081920 minix
010:  -----  0001173504 0001999999 0000826496 Unallocated

```

Nie ma takiego obszaru jak gpt_load_table. Jeśli mowa o GPT Header to jego sektor startowy to 1, a Partition Table zaczyna się w sektorze 2. W obrazie znajduje się 826496+2048 niezaalokowanych sektorów. Ich start i koniec to: 0001173504 i 001999999 oraz 0 i 2047.

Ujawnione woluminy są widoczne na rzucie ekranu (skrajna prawa kolumna).

Pomimo prób, sukcesu br

Nie byłem w stanie wyświetlić informacji o tablicy partycji z wykorzystaniem mmstats.

Pomimo wielu prób z wykorzystaniem różnych offsetów.

```
(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5848783 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 0 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5848783 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 1 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5848783 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 3 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5831774 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 4 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5823387 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 32 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5814883 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmstat -o 33 ./LAB_1.img
gpt
completion in 27 second(s) with 5.5 MiB/s (5814883 bytes/second).

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ mmls ./LAB_1.img
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Safety Table
001:	-----	0000000000	0000002047	0000002048	Unallocated
002:	Meta	0000000001	0000000001	0000000001	GPT Header
003:	Meta	0000000002	0000000033	0000000032	Partition Table
004:	000	0000002048	0000104447	0000102400	fat16
005:	001	0000104448	0000309247	0000204800	fat32
006:	002	0000309248	0000718847	0000409600	ntfs
007:	003	0000718848	0001058815	0000339968	ext4
008:	004	0001058816	0001091583	0000032768	swap
009:	005	0001091584	0001173503	0000081920	minix
010:	-----	0001173504	0001999999	0000826496	Unallocated

Volume Serial Number:
451AF24C771A6637
Version: Windows XP

```
uzytkownik1@uzytkownik1: ~/Downloads
010: ----- 00011/3504 0001999999 0000826496 Unallocated
acquired 15.000 (15.000 MiB/s) with 5.0 MiB/s (5848783 bytes/second).
(uzytkownik1@uzytkownik1) [~/Downloads]
$ fsstat -o 309248 ./LAB_1.img
FILE SYSTEM INFORMATION
-----
File System Type: NTFS (ondisk) with 5.0 MiB/s (5848783 bytes/second).
Volume Serial Number: 451AF24C771A6637
File Name: NTFS
Volume Name: NTFS (13939371808 bytes) of total: 16 GiB (16843212800 B)
Version: Windows XP (second(s) with 5.0 MiB/s (5848783 bytes/second)).

METADATA INFORMATION
-----
First Cluster of MFT: 4 (ondisk) with 5.0 MiB/s (5848783 bytes/second).
First Cluster of MFT Mirror: 25599
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes (bytes) of total: 16 GiB (16843212800 B)
Range: 0 - 68 (10.22 second(s) with 5.0 MiB/s (5848783 bytes/second)).
Root Directory: 5

CONTENT INFORMATION (16008879168 bytes) of total: 16 GiB (16843212800 B)
----- (5848783 bytes/second).
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 51198 (bytes) of total: 16 GiB (16843212800 B)
Total Sector Range: 0 - 409598 with 5.0 MiB/s (5848783 bytes/second).

AttrDef Attribute Values: 1024 00:00:00
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident (s) with
$FILE_NAME (48) Size: 68-578 Flags: Resident, Index (00:00:00)
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
```

Kopiowanie dysku

```
(uzytkownik1@uzytkownik1)-[~]
$ sudo fdisk -l
```

Disk /dev/sda: 93.4 GiB, 100288954368 bytes, 195876864 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe8776c52

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1	*	2048	193875967	193873920	92.4G	83	Linux
/dev/sda2		193878014	195874815	1996802	975M	f	W95 Ext'd (LBA)
/dev/sda5		193878016	195874815	1996800	975M	82	Linux swap / Solaris

Disk /dev/sdb: 14.94 GiB, 16043212800 bytes, 31334400 sectors
Disk model: Flash Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcbfe1303

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1	*	2048	31334399	31332352	14.9G	c	W95 FAT32 (LBA)

```
(uzytkownik1@uzytkownik1)-[~]
$ s
```

Status: at 99%.

Volume acquired 14 GiB (15939371008 bytes) of total 14 GiB (16043212800 bytes).
Volume completion in 27 second(s) with 5.5 MiB/s (5848783 bytes/second).

Status: at 99%.

File acquired 14 GiB (15962177536 bytes) of total 14 GiB (16043212800 bytes).
File completion in 27 second(s) with 5.5 MiB/s (5840266 bytes/second).

First Cluster of MFT Mirror: 20099

Status: at 99%.

Size acquired 14 GiB (15985508352 bytes) of total 14 GiB (16043212800 bytes).
Range completion in 27 second(s) with 5.5 MiB/s (5831774 bytes/second).

Root Directory: 0

Status: at 99%.

COM acquired 14 GiB (16008839168 bytes) of total 14 GiB (16043212800 bytes).
completion in 27 second(s) with 5.5 MiB/s (5823307 bytes/second).

Section Size: 512

Status: at 99%.

Total acquired 14 GiB (16032169984 bytes) of total 14 GiB (16043212800 bytes).
Total completion in 27 second(s) with 5.5 MiB/s (5814865 bytes/second).

Acquiry completed at: Oct 23, 2024 00:03:03

STANDARD INFORMATION (16) Size: 48-72 Flags: Resident

Written: 14 GiB (16043212988 bytes) in 45 minute(s) and 33 second(s) with 5.6 MiB/s (5870184 bytes/second).

MD5 hash calculated over data: 378 FL 542d580ccaf35ada3f09c954f806eb04

ewfacquire: SUCCESS Size: 0-256 Flags: Resident

SECURITY INFORMATION (80) Size: No Limit Flags: Non-Resident

(uzytkownik1@uzytkownik1)-[~] Size: No Limit Flags: Resident

\$ S LUM INFORMATION (112) Size: 12-12 Flags: Resident

Weryfikacja kopiowania nośnika

Przebiegła pomyślnie :)

Dziękuję za uwagę i przepraszam za opóźnienie w oddaniu pracy. Mam nadzieję, że nie będzie to duży problem, z uwagi na to, że i tak są to godziny nocne.

```
uzytkownik1@uzytkownik1: ~/Downloads

(uzytkownik1@uzytkownik1)-[~/Downloads]
$ ewfverify ./USB_4GB_Kingston.E01.E01
ewfverify 20140814

Verify started at: Oct 23, 2024 00:23:15
This could take a while.

Status: at 11%.
verified 1.7 GiB (1872494592 bytes) of total 14 GiB (16043212800 bytes).
completion in 32 second(s) with 425 MiB/s (445644800 bytes/second).

Status: at 25%.
verified 3.8 GiB (4097376256 bytes) of total 14 GiB (16043212800 bytes).
completion in 24 second(s) with 478 MiB/s (501350400 bytes/second).

Status: at 38%.
verified 5.7 GiB (6178897920 bytes) of total 14 GiB (16043212800 bytes).
completion in 19 second(s) with 493 MiB/s (517522993 bytes/second).

Status: at 51%.
verified 7.7 GiB (8291024896 bytes) of total 14 GiB (16043212800 bytes).
completion in 15 second(s) with 493 MiB/s (517522993 bytes/second).

Status: at 64%.
verified 9.6 GiB (10332110848 bytes) of total 14 GiB (16043212800 bytes).
completion in 11 second(s) with 493 MiB/s (517522993 bytes/second).

Status: at 77%.
verified 11 GiB (12423888896 bytes) of total 14 GiB (16043212800 bytes).
completion in 7 second(s) with 493 MiB/s (517522993 bytes/second).

Status: at 90%.
verified 13 GiB (14563573760 bytes) of total 14 GiB (16043212800 bytes).
completion in 3 second(s) with 493 MiB/s (517522993 bytes/second).

Verify completed at: Oct 23, 2024 00:23:45

Read: 14 GiB (16043212800 bytes) in 30 second(s) with 510 MiB/s (534773760 bytes/second).

MD5 hash stored in file: 542d580ccaf35ada3f09c954f806eb04
MD5 hash calculated over data: 542d580ccaf35ada3f09c954f806eb04

ewfverify: SUCCESS

(uzytkownik1@uzytkownik1)-[~/Downloads]
$
```